

OOPS ASSIGNMENT – 06

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1.

```
#include <iostream>
```

```
using namespace std;
```

```
class Book
```

```
{
```

```
public:
```

```
    string title, author, ISBN;
```

```
};
```

```
class Library
```

```
{
```

```
public:
```

```
    Book b[10];
```

```
    int count = 0;
```

```
    bool addNewBook(string &title, string &author, string &ISBN)
```

```
{
```

```
    if (count >= 10)
```

```
        return false;
```

```
    b[count].title = title;
```

```
    b[count].author = author;
```

```
    b[count].ISBN = ISBN;
```

```
    count++;  
    return true;  
}
```

```
bool removeBooks(string &ISBN);
```

```
void displayDetails()  
{  
    for (int i = 0; i < count; i++)  
    {  
        cout << b[i].title << " " << b[i].author << " " << b[i].ISBN << endl;  
    }  
}  
};
```

```
bool Library::removeBooks(string &ISBN)
```

```
{  
    for (int i = 0; i < count; i++)  
    {  
        if (b[i].ISBN == ISBN)  
        {  
            for (int j = i; j < count - 1; j++)  
            {  
                b[j] = b[j + 1];  
            }  
            count--;  
            return true;  
        }  
    }  
}
```

```

    }
}
return false;
}

int main()
{
    Library l;

    string t, a, i;

    t="C++"; a="Bjarne"; i="1"; l.addNewBook(t,a,i);
    t="Java"; a="James"; i="2"; l.addNewBook(t,a,i);
    t="Python"; a="Guido"; i="3"; l.addNewBook(t,a,i);
    t="DSA"; a="CLRS"; i="4"; l.addNewBook(t,a,i);
    t="OS"; a="Galvin"; i="5"; l.addNewBook(t,a,i);

    string rem = "3";

    l.removeBooks(rem);

    l.displayDetails();

    return 0;
}

```

```

● (base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash
C++ Bjarne 1
Java James 2
DSA CLRS 4
OS Galvin 5
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```

2.

```
#include <iostream>
```

```
using namespace std;
```

```
class Book
```

```
{
```

```
public:
```

```
    string title, author, ISBN;
```

```
    Book()
```

```
{
```

```
    title = "";
```

```
    author = "";
```

```
    ISBN = "";
```

```
}
```

```
    Book(string title, string author, string ISBN)
```

```
{
```

```
    this->title = title;
```

```
    this->author = author;
```

```
    this->ISBN = ISBN;
```

```
}
```

```
    Book(const Book &b)
```

```
{
```

```
    this->title = b.title;
```

```
    this->author = b.author;
```

```
    this->ISBN = b.ISBN;
```

```
    }  
};
```

```
class Library
```

```
{
```

```
public:
```

```
    Book b[10];
```

```
    int count = 0;
```

```
    bool addNewBook(string &title, string &author, string &ISBN)
```

```
{
```

```
    if (count >= 10)
```

```
        return false;
```

```
    b[count] = Book(title, author, ISBN);
```

```
    count++;
```

```
    return true;
```

```
}
```

```
    bool removeBooks(string &ISBN)
```

```
{
```

```
    for (int i = 0; i < count; i++)
```

```
{
```

```
    if (b[i].ISBN == ISBN)
```

```
{
```

```
        for (int j = i; j < count - 1; j++)
```

```
            b[j] = b[j + 1];
```

```
        count--;  
        return true;  
    }  
}  
return false;  
}
```

```
void displayDetails()  
{  
    for (int i = 0; i < count; i++)  
    {  
        cout << b[i].title << " " << b[i].author << " " << b[i].ISBN << endl;  
    }  
}  
};
```

```
int main()  
{  
    Library l;  
  
    string t,a,i;  
  
    t="C++"; a="Bjarne"; i="1"; l.addNewBook(t,a,i);  
    t="Java"; a="James"; i="2"; l.addNewBook(t,a,i);  
    t="Python"; a="Guido"; i="3"; l.addNewBook(t,a,i);
```

```
    Book arr[2] = {Book("DSA","CLRS","4"), Book("OS","Galvin","5")};
```

```
for(int j=0;j<2;j++)  
    l.addNewBook(arr[j].title, arr[j].author, arr[j].ISBN);
```

```
string rem="3";  
l.removeBooks(rem);
```

```
l.displayDetails();
```

```
return 0;
```

```
}
```

```
● (base) vihaan@Vihaans-Macbook 10.GRAPHs % cd "/Users/vihaan/Desktop/DSA/10.GRAPHs/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPHs/"trash  
C++ Bjarne 1  
Java James 2  
DSA CLRS 4  
OS Galvin 5  
○ (base) vihaan@Vihaans-Macbook 10.GRAPHs %
```

3.

```
#include <iostream>  
using namespace std;
```

```
class Account
```

```
{
```

```
public:
```

```
    const long acc_no;
```

```
    long trans_id;
```

```
    string type;
```

```
    double balance;
```

```
    Account(long a, double b) : acc_no(a)
```

```
{
```

```
    balance = b;  
    trans_id = 0;  
    type = "";  
}
```

```
long depositAmount(const long to, const long from, const double amount)  
{  
    if (to == acc_no)  
    {  
        balance += amount;  
        trans_id++;  
        type = "credit";  
    }  
    return trans_id;  
}
```

```
long creditAmount(const long to, const long from, const double amount)  
{  
    if (from == acc_no)  
    {  
        balance -= amount;  
        trans_id++;  
        type = "debit";  
    }  
    return trans_id;  
}
```



```

void displayDetails() const
{
    cout << acc_no << " " << balance << " " << type << endl;
}

};

int main()
{
    Account a1(1,1000), a2(2,2000), a3(3,3000), a4(4,4000), a5(5,5000);

    a1.depositAmount(1,2,500);
    a2.creditAmount(3,2,200);
    a3.depositAmount(3,1,300);
    a4.creditAmount(5,4,100);
    a5.depositAmount(5,2,700);

    a1.displayDetails();
    a2.displayDetails();
    a3.displayDetails();
    a4.displayDetails();
    a5.displayDetails();

    return 0;
}

```

```

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1 1500 credit
2 1800 debit
3 3300 credit
4 3900 debit
5 5700 credit
(base) vihaan@Vihaans-Macbook 10.GRAPH5 %

```

4.

```
#include <iostream>

using namespace std;
```

```
class B;
```

```
class A
```

```
{
```

```
public:
```

```
    int x;
```

```
    A(int a)
```

```
{
```

```
    x = a;
```

```
}
```

```
    friend int add(A, B);
```

```
};
```

```
class B
```

```
{
```

```
public:
```

```
    int y;
```

```
    B(int b)
```

```
{
```

```
    y = b;
```

```
}
```

```
friend int add(A, B);  
};
```

```
int add(A a, B b)  
{  
    return a.x + b.y;  
}
```

```
int main()  
{  
    A a(5);  
    B b(10);  
  
    cout << add(a, b);  
  
    return 0;  
}
```

```
● (base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash  
15  
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```

5.

```
#include <iostream>  
using namespace std;
```

```
class Complex
```

```
{
```

```
public:
```

```
    int real, imag;
```

```
Complex(int r, int i)
```

```
{  
    real = r;  
    imag = i;  
}
```

```
Complex(const Complex &c)
```

```
{  
    real = c.real;  
    imag = c.imag;  
}
```

```
void display()
```

```
{  
    cout << real << " + " << imag << "i" << endl;  
}
```

```
friend void sum(Complex, Complex);
```

```
};
```

```
void sum(Complex c1, Complex c2)
```

```
{  
    Complex c3(0,0);  
    c3.real = c1.real + c2.real;  
    c3.imag = c1.imag + c2.imag;  
    c3.display();  
}
```

```
}
```

```
int main()
```

```
{
```

```
    Complex c1(2,3);
```

```
    Complex c2(4,5);
```

```
    Complex c3 = c1;
```

```
    c1.display();
```

```
    c2.display();
```

```
    c3.display();
```

```
    sum(c1, c2);
```

```
    return 0;
```

```
}
```

```
● (base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash
2 + 3i
4 + 5i
2 + 3i
6 + 8i
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```